

Improving Language Understanding by Generative Pre-Training

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Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.¹ Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

¹<https://gluebenchmark.com/leaderboard>

通过生成式预训练提升语言理解能力

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摘要

自然语言理解涵盖了广泛的多样化任务，如文本蕴含、问答、语义相似性评估和文档分类。尽管大规模未标注文本语料库丰富，但用于学习这些特定任务的标注数据却稀缺，这使得判别式训练模型难以充分表现。我们证明，通过在多样化的未标注文本语料库上*generative pre-training*语言模型，随后在每个特定任务上进行*discriminative fine-tuning*，可以在这些任务上实现显著提升。与先前方法不同，我们在微调过程中利用任务感知的输入转换，以实现有效迁移，同时只需对模型架构进行最小改动。我们在自然语言理解的广泛基准测试中展示了我们方法的有效性。我们的通用任务无关模型优于为每个任务专门设计的判别式训练模型，在研究的12项任务中有9项显著提升了现有技术水平。例如，我们在常识推理（Stories Cloze Test）上实现了8.9%的绝对提升，在问答（RACE）上实现了5.7%的提升，在文本蕴含（MultiNLI）上实现了1.5%的提升。

1 引言

从原始文本中有效学习的能力对于减轻自然语言处理（NLP）中对监督学习的依赖至关重要。大多数深度学习方法需要大量手动标注的数据，这限制了它们在许多标注资源匮乏领域的应用[61]。在这些情况下，能够利用未标注数据中语言信息的模型为收集更多标注提供了一种有价值的替代方案，因为后者可能耗时且昂贵。此外，即使在有大量监督数据可用的情况下，以无监督方式学习良好的表示也能显著提升性能。迄今为止，最有力的证据是预训练词嵌入[10, 39, 42]的广泛使用，它提升了一系列NLP任务的性能[8, 11, 26, 45]。

然而，从未标注文本中利用超越词汇层面的信息，主要面临两大挑战。首先，尚不清楚何种优化目标最有利于学习适用于迁移的文本表示。近期研究探讨了多种目标，如语言建模[44]、机器翻译[38]和语篇连贯性[22]，每种方法在不同任务中各有所长。¹其次，对于如何将这些学习到的表示最有效地迁移至目标任务，学界尚未达成共识。现有技术涉及对模型架构进行任务特定调整[43, 44]、采用复杂学习方案[21]以及添加辅助学习目标[50]等组合策略。这些不确定性使得开发有效的语言处理半监督学习方法变得困难。

¹<https://gluebenchmark.com/leaderboard>

In this paper, we explore a semi-supervised approach for language understanding tasks using a combination of unsupervised pre-training and supervised fine-tuning. Our goal is to learn a universal representation that transfers with little adaptation to a wide range of tasks. We assume access to a large corpus of unlabeled text and several datasets with manually annotated training examples (target tasks). Our setup does not require these target tasks to be in the same domain as the unlabeled corpus. We employ a two-stage training procedure. First, we use a language modeling objective on the unlabeled data to learn the initial parameters of a neural network model. Subsequently, we adapt these parameters to a target task using the corresponding supervised objective.

For our model architecture, we use the *Transformer* [62], which has been shown to perform strongly on various tasks such as machine translation [62], document generation [34], and syntactic parsing [29]. This model choice provides us with a more structured memory for handling long-term dependencies in text, compared to alternatives like recurrent networks, resulting in robust transfer performance across diverse tasks. During transfer, we utilize task-specific input adaptations derived from traversal-style approaches [52], which process structured text input as a single contiguous sequence of tokens. As we demonstrate in our experiments, these adaptations enable us to fine-tune effectively with minimal changes to the architecture of the pre-trained model.

We evaluate our approach on four types of language understanding tasks – natural language inference, question answering, semantic similarity, and text classification. Our general task-agnostic model outperforms discriminatively trained models that employ architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test) [40], 5.7% on question answering (RACE) [30], 1.5% on textual entailment (MultiNLI) [66] and 5.5% on the recently introduced GLUE multi-task benchmark [64]. We also analyzed zero-shot behaviors of the pre-trained model on four different settings and demonstrate that it acquires useful linguistic knowledge for downstream tasks.

2 Related Work

Semi-supervised learning for NLP Our work broadly falls under the category of semi-supervised learning for natural language. This paradigm has attracted significant interest, with applications to tasks like sequence labeling [24, 33, 57] or text classification [41, 70]. The earliest approaches used unlabeled data to compute word-level or phrase-level statistics, which were then used as features in a supervised model [33]. Over the last few years, researchers have demonstrated the benefits of using word embeddings [11, 39, 42], which are trained on unlabeled corpora, to improve performance on a variety of tasks [8, 11, 26, 45]. These approaches, however, mainly transfer word-level information, whereas we aim to capture higher-level semantics.

Recent approaches have investigated learning and utilizing more than word-level semantics from unlabeled data. Phrase-level or sentence-level embeddings, which can be trained using an unlabeled corpus, have been used to encode text into suitable vector representations for various target tasks [28, 32, 1, 36, 22, 12, 56, 31].

Unsupervised pre-training Unsupervised pre-training is a special case of semi-supervised learning where the goal is to find a good initialization point instead of modifying the supervised learning objective. Early works explored the use of the technique in image classification [20, 49, 63] and regression tasks [3]. Subsequent research [15] demonstrated that pre-training acts as a regularization scheme, enabling better generalization in deep neural networks. In recent work, the method has been used to help train deep neural networks on various tasks like image classification [69], speech recognition [68], entity disambiguation [17] and machine translation [48].

The closest line of work to ours involves pre-training a neural network using a language modeling objective and then fine-tuning it on a target task with supervision. Dai et al. [13] and Howard and Ruder [21] follow this method to improve text classification. However, although the pre-training phase helps capture some linguistic information, their usage of LSTM models restricts their prediction ability to a short range. In contrast, our choice of transformer networks allows us to capture longer-range linguistic structure, as demonstrated in our experiments. Further, we also demonstrate the effectiveness of our model on a wider range of tasks including natural language inference, paraphrase detection and story completion. Other approaches [43, 44, 38] use hidden representations from a

在本文中，我们探索了一种结合无监督预训练与有监督微调的半监督方法，用于语言理解任务。我们的目标是学习一种通用表示，该表示只需少量调整即可迁移至广泛的任务中。我们假设可以访问大量未标注文本语料库以及多个包含人工标注训练样本的数据集（目标任务）。我们的设置不要求这些目标任务与未标注语料库处于同一领域。我们采用两阶段训练流程：首先，在未标注数据上使用语言建模目标来学习神经网络模型的初始参数；随后，利用相应的有监督目标将这些参数适配到目标任务中。

在我们的模型架构中，我们采用了*Transformer* [62]，该模型已在机器翻译[62]、文档生成[34]和句法分析[29]等多种任务中展现出强大的性能。与循环网络等替代方案相比，这一模型选择为我们提供了更结构化的记忆机制，以处理文本中的长期依赖关系，从而在不同任务间实现稳健的迁移性能。在迁移过程中，我们采用了基于遍历式方法[52]的任务特定输入适配技术，该技术将结构化文本输入作为连续的单一标记序列进行处理。正如我们在实验中所展示的，这些适配方法使我们能够在对预训练模型架构进行最小改动的情況下，实现有效的微调。

我们在四种类型的语言理解任务上评估了我们的方法——自然语言推理、问答、语义相似度和文本分类。我们的通用任务无关模型超越了那些为每个任务专门设计架构的判别式训练模型，在研究的12项任务中有9项显著提升了现有技术水平。例如，我们在常识推理（Stories Cloze Test）[40]上实现了8.9%的绝对提升，在问答（RACE）[30]上实现了5.7%的提升，在文本蕴含（MultiNLI）[66]上实现了1.5%的提升，以及在最近推出的GLUE多任务基准测试[64]上实现了5.5%的提升。我们还分析了预训练模型在四种不同设置下的零样本行为，并证明它获得了对下游任务有用的语言学知识。

2 相关工作

自然语言处理的半监督学习我们的工作大致属于自然语言半监督学习的范畴。这一范式已引起广泛关注，应用于序列标注[24, 33, 57]或文本分类[41, 70]等任务。最早的方法使用未标注数据计算词级或短语级统计信息，随后将其作为监督模型的特征[33]。过去几年中，研究者证明了利用在未标注语料库上训练的词嵌入[11, 39, 42]能够提升多种任务的性能[8, 11, 26, 45]。然而这些方法主要传递词级信息，而我们的目标是捕捉更高层次的语义。

最近的研究方法探索了从无标注数据中学习和利用超越词汇层面的语义。通过无标注语料库训练得到的短语级或句子级嵌入，已被用于将文本编码成适合各种目标任务[28, 32, 1, 36, 22, 12, 56, 31]的向量表示。

无监督预训练 无监督预训练是半监督学习的一种特殊形式，其目标是寻找一个良好的初始点，而非修改监督学习的目标。早期研究探索了该技术在图像分类[20, 49, 63]和回归任务[3]中的应用。后续研究[15]表明，预训练作为一种正则化方案，能够在深度神经网络中实现更好的泛化。在近期工作中，该方法已被用于辅助训练深度神经网络处理各类任务，如图像分类[69]、语音识别[68]、实体消歧[17]和机器翻译[48]。

与我们工作最接近的研究方向，涉及使用语言建模目标预训练神经网络，然后在监督下针对目标任务进行微调。Dai等人[13]以及Howard和Ruder[21]采用这种方法改进了文本分类。然而，尽管预训练阶段有助于捕捉某些语言信息，但他们使用的LSTM模型将其预测能力限制在较短范围内。相比之下，我们选择的transformer网络能够捕捉更长距离的语言结构，正如我们实验所展示的。此外，我们还证明了我们的模型在更广泛任务上的有效性，包括自然语言推理、复述检测和故事补全。其他方法[43, 44, 38]则利用来自

pre-trained language or machine translation model as auxiliary features while training a supervised model on the target task. This involves a substantial amount of new parameters for each separate target task, whereas we require minimal changes to our model architecture during transfer.

Auxiliary training objectives Adding auxiliary unsupervised training objectives is an alternative form of semi-supervised learning. Early work by Collobert and Weston [10] used a wide variety of auxiliary NLP tasks such as POS tagging, chunking, named entity recognition, and language modeling to improve semantic role labeling. More recently, Rei [50] added an auxiliary language modeling objective to their target task objective and demonstrated performance gains on sequence labeling tasks. Our experiments also use an auxiliary objective, but as we show, unsupervised pre-training already learns several linguistic aspects relevant to target tasks.

3 Framework

Our training procedure consists of two stages. The first stage is learning a high-capacity language model on a large corpus of text. This is followed by a fine-tuning stage, where we adapt the model to a discriminative task with labeled data.

3.1 Unsupervised pre-training

Given an unsupervised corpus of tokens $\mathcal{U} = \{u_1, \dots, u_n\}$, we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

where k is the size of the context window, and the conditional probability P is modeled using a neural network with parameters Θ . These parameters are trained using stochastic gradient descent [51].

In our experiments, we use a multi-layer *Transformer decoder* [34] for the language model, which is a variant of the transformer [62]. This model applies a multi-headed self-attention operation over the input context tokens followed by position-wise feedforward layers to produce an output distribution over target tokens:

$$\begin{aligned} h_0 &= UW_e + W_p \\ h_l &= \text{transformer_block}(h_{l-1}) \forall l \in [1, n] \\ P(u) &= \text{softmax}(h_n W_e^T) \end{aligned} \quad (2)$$

where $U = (u_{-k}, \dots, u_{-1})$ is the context vector of tokens, n is the number of layers, W_e is the token embedding matrix, and W_p is the position embedding matrix.

3.2 Supervised fine-tuning

After training the model with the objective in Eq. 1, we adapt the parameters to the supervised target task. We assume a labeled dataset $\mathcal{C}$, where each instance consists of a sequence of input tokens, $x^1, \dots, x^m$, along with a label y . The inputs are passed through our pre-trained model to obtain the final transformer block’s activation h_l^m , which is then fed into an added linear output layer with parameters W_y to predict y :

$$P(y | x^1, \dots, x^m) = \text{softmax}(h_l^m W_y). \quad (3)$$

This gives us the following objective to maximize:

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y | x^1, \dots, x^m). \quad (4)$$

We additionally found that including language modeling as an auxiliary objective to the fine-tuning helped learning by (a) improving generalization of the supervised model, and (b) accelerating convergence. This is in line with prior work [50, 43], who also observed improved performance with such an auxiliary objective. Specifically, we optimize the following objective (with weight λ):

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \quad (5)$$

Overall, the only extra parameters we require during fine-tuning are W_y , and embeddings for delimiter tokens (described below in Section 3.3).

在目标任务上训练监督模型时，使用预训练语言或机器翻译模型作为辅助特征。这需要为每个独立的目标任务引入大量新参数，而我们在迁移过程中要求对模型架构进行最小改动。

辅助训练目标 添加辅助无监督训练目标是半监督学习的另一种形式。Collobert和Weston的早期研究[10]使用了多种辅助自然语言处理任务，如词性标注、组块分析、命名实体识别和语言建模，以改进语义角色标注。最近，Rei[50]在其目标任务目标中增加了辅助语言建模目标，并在序列标注任务上展示了性能提升。我们的实验也使用了辅助目标，但正如我们所展示的，无监督预训练已经学习了与目标任务相关的多个语言方面。

3 框架

我们的训练过程包括两个阶段。第一阶段是在大量文本语料上学习一个高容量的语言模型。随后是微调阶段，在此阶段中，我们使用标注数据将模型适配到判别性任务上。

3.1 无监督预训练

给定一个无监督的词汇标记集合 $\mathcal{U} = \{u_1, \dots, u_n\}$ ，我们采用标准的语言建模目标来最大化以下似然函数：

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

其中 k 是上下文窗口的大小，条件概率 P 通过一个参数为 Θ 的神经网络建模。这些参数使用随机梯度下降法进行训练[51]。

在我们的实验中，我们使用了一个多层 *Transformer decoder* [34] 作为语言模型，这是 transformer [62] 的一个变体。该模型在输入上下文标记上应用多头自注意力操作，随后通过位置前馈层生成目标标记的输出分布：

$$\begin{aligned} h_0 &= UW_e + W_p \\ h_l &= \text{transformer_block}(h_{l-1}) \forall l \in [1, n] \\ P(u) &= \text{softmax}(h_n W_e^T) \end{aligned} \quad (2)$$

其中 $U = (u_{-k}, \dots, u_{-1})$ 是词元的上下文向量， n 是层数， W_e 是词元嵌入矩阵， W_p 是位置嵌入矩阵。

3.2 监督微调

在使用公式1中的目标训练模型后，我们调整参数以适应有监督的目标任务。假设有一个带标签的数据集 $\mathcal{C}$ ，其中每个实例由一系列输入标记 $x^1, \dots, x^m$ 和一个标签 y 组成。将这些输入通过我们预训练的模型，以获得最终Transformer块的激活值 h_l^m ，然后将其输入到一个新增的线性输出层，该层参数为 W_y ，用于预测 y ：

$$P(y | x^1, \dots, x^m) = \text{softmax}(h_l^m W_y). \quad (3)$$

这为我们提供了以下需要最大化的目标：

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y | x^1, \dots, x^m). \quad (4)$$

此外，我们发现将语言建模作为微调的辅助目标有助于学习，具体体现在：(a) 提升监督模型的泛化能力，(b) 加速收敛过程。这与先前的研究[50, 43]结论一致，这些研究同样观察到此类辅助目标带来的性能提升。具体而言，我们优化了以下目标函数（权重为 λ ）：

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \quad (5)$$

总的来说，在微调过程中，我们唯一需要的额外参数是 W_y ，以及分隔符标记的嵌入（在下面的3.3节中描述）。

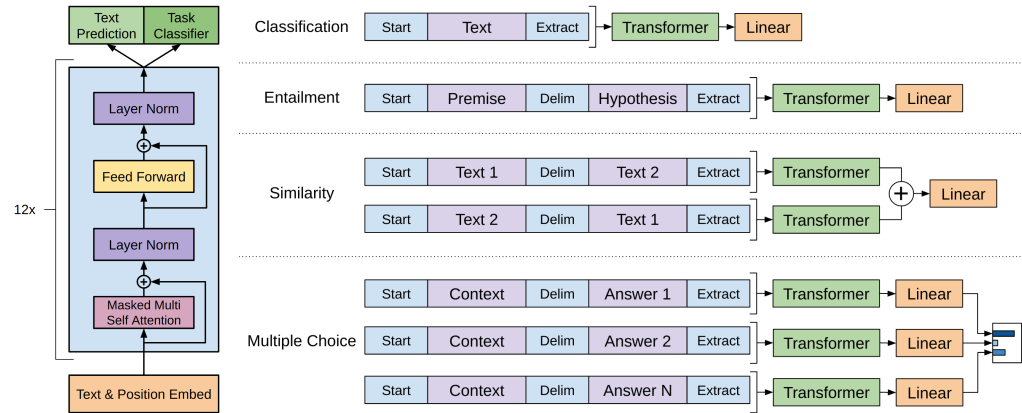


Figure 1: **(left)** Transformer architecture and training objectives used in this work. **(right)** Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

3.3 Task-specific input transformations

For some tasks, like text classification, we can directly fine-tune our model as described above. Certain other tasks, like question answering or textual entailment, have structured inputs such as ordered sentence pairs, or triplets of document, question, and answers. Since our pre-trained model was trained on contiguous sequences of text, we require some modifications to apply it to these tasks. Previous work proposed learning task specific architectures on top of transferred representations [44]. Such an approach re-introduces a significant amount of task-specific customization and does not use transfer learning for these additional architectural components. Instead, we use a traversal-style approach [52], where we convert structured inputs into an ordered sequence that our pre-trained model can process. These input transformations allow us to avoid making extensive changes to the architecture across tasks. We provide a brief description of these input transformations below and Figure 1 provides a visual illustration. All transformations include adding randomly initialized start and end tokens ($\langle s \rangle$, $\langle e \rangle$).

Textual entailment For entailment tasks, we concatenate the premise p and hypothesis h token sequences, with a delimiter token ($\$$) in between.

Similarity For similarity tasks, there is no inherent ordering of the two sentences being compared. To reflect this, we modify the input sequence to contain both possible sentence orderings (with a delimiter in between) and process each independently to produce two sequence representations h_l^m which are added element-wise before being fed into the linear output layer.

Question Answering and Commonsense Reasoning For these tasks, we are given a context document z , a question q , and a set of possible answers $\{a_k\}$. We concatenate the document context and question with each possible answer, adding a delimiter token in between to get $[z; q; \$; a_k]$. Each of these sequences are processed independently with our model and then normalized via a softmax layer to produce an output distribution over possible answers.

4 Experiments

4.1 Setup

Unsupervised pre-training We use the BooksCorpus dataset [71] for training the language model. It contains over 7,000 unique unpublished books from a variety of genres including Adventure, Fantasy, and Romance. Crucially, it contains long stretches of contiguous text, which allows the generative model to learn to condition on long-range information. An alternative dataset, the 1B Word Benchmark, which is used by a similar approach, ELMo [44], is approximately the same size

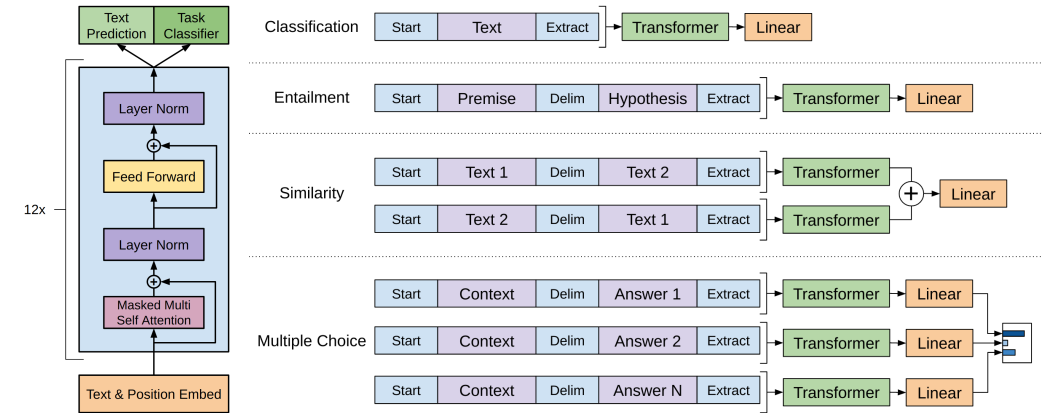


图1: (左) 本研究使用的Transformer架构与训练目标。(右) 针对不同任务进行微调的输入转换方法。我们将所有结构化输入转换为词元序列, 通过预训练模型处理后, 接入线性+softmax层。

3.3 任务特定输入转换

对于某些任务, 如文本分类, 我们可以直接按照上述方法对模型进行微调。而其他一些任务, 如问答或文本蕴含, 具有结构化的输入, 例如有序的句子对, 或文档、问题和答案的三元组。由于我们的预训练模型是在连续的文本序列上训练的, 因此需要对其进行一些修改才能应用于这些任务。先前的研究提出了在迁移表示之上学习特定任务的架构[44]。这种方法重新引入了大量特定于任务的定制, 并且没有对这些额外的架构组件使用迁移学习。相反, 我们采用了一种遍历式方法[52], 将结构化输入转换为我们的预训练模型可以处理的有序序列。这些输入转换使我们能够避免在不同任务中对架构进行大量修改。下面我们简要描述这些输入转换, 图1提供了直观的图示。所有转换都包括添加随机初始化的起始和结束标记 ($\langle s \rangle$, $\langle e \rangle$)。

文本蕴含 对于蕴含任务, 我们将前提 p 和假设 h 的标记序列连接起来, 中间用分隔符 ($\$$) 隔开。

相似性 对于相似性任务, 被比较的两个句子没有固有的顺序。为了反映这一点, 我们修改输入序列, 使其包含两种可能的句子顺序 (中间用分隔符隔开), 并独立处理每个顺序, 生成两个序列表示 h_l^m , 这两个表示在送入线性输出层之前会进行逐元素相加。

问答与常识推理 对于这些任务, 我们会获得一个上下文文档 一个问题 q 以及一组可能的答案 $\{a_k\}$ 。我们将文档上下文和问题与每个可能的答案拼接起来, 在中间添加一个分隔符标记, 得到 $[z; q; \$; a_k]$ 。这些序列中的每一个都通过我们的模型独立处理, 然后通过 softmax 层进行归一化, 以生成可能答案的输出分布。

4 实验

4.1 设置

无监督预训练 我们使用BooksCorpus数据集[71]来训练语言模型。该数据集包含超过7,000本独特的未出版书籍, 涵盖冒险、奇幻和爱情等多种体裁。关键在于它包含大量连续文本段落, 这使得生成模型能够学习基于长距离信息的条件生成。类似方法ELMo[44]采用的替代数据集——1B Word Benchmark——规模与之大致相当。

Table 1: A list of the different tasks and datasets used in our experiments.

Task	Datasets
Natural language inference	SNLI [5], MultiNLI [66], Question NLI [64], RTE [4], SciTail [25]
Question Answering	RACE [30], Story Cloze [40]
Sentence similarity	MSR Paraphrase Corpus [14], Quora Question Pairs [9], STS Benchmark [6]
Classification	Stanford Sentiment Treebank-2 [54], CoLA [65]

but is shuffled at a sentence level - destroying long-range structure. Our language model achieves a very low token level perplexity of 18.4 on this corpus.

Model specifications Our model largely follows the original transformer work [62]. We trained a 12-layer decoder-only transformer with masked self-attention heads (768 dimensional states and 12 attention heads). For the position-wise feed-forward networks, we used 3072 dimensional inner states. We used the Adam optimization scheme [27] with a max learning rate of 2.5e-4. The learning rate was increased linearly from zero over the first 2000 updates and annealed to 0 using a cosine schedule. We train for 100 epochs on minibatches of 64 randomly sampled, contiguous sequences of 512 tokens. Since layernorm [2] is used extensively throughout the model, a simple weight initialization of $N(0, 0.02)$ was sufficient. We used a bytetrain encoding (BPE) vocabulary with 40,000 merges [53] and residual, embedding, and attention dropouts with a rate of 0.1 for regularization. We also employed a modified version of L2 regularization proposed in [37], with $w = 0.01$ on all non bias or gain weights. For the activation function, we used the Gaussian Error Linear Unit (GELU) [18]. We used learned position embeddings instead of the sinusoidal version proposed in the original work. We use the *ftfy* library² to clean the raw text in BooksCorpus, standardize some punctuation and whitespace, and use the *spaCy* tokenizer.³

Fine-tuning details Unless specified, we reuse the hyperparameter settings from unsupervised pre-training. We add dropout to the classifier with a rate of 0.1. For most tasks, we use a learning rate of 6.25e-5 and a batchsize of 32. Our model finetunes quickly and 3 epochs of training was sufficient for most cases. We use a linear learning rate decay schedule with warmup over 0.2% of training. λ was set to 0.5.

4.2 Supervised fine-tuning

We perform experiments on a variety of supervised tasks including natural language inference, question answering, semantic similarity, and text classification. Some of these tasks are available as part of the recently released GLUE multi-task benchmark [64], which we make use of. Figure 1 provides an overview of all the tasks and datasets.

Natural Language Inference The task of natural language inference (NLI), also known as recognizing textual entailment, involves reading a pair of sentences and judging the relationship between them from one of `entailment`, `contradiction` or `neutral`. Although there has been a lot of recent interest [58, 35, 44], the task remains challenging due to the presence of a wide variety of phenomena like lexical entailment, coreference, and lexical and syntactic ambiguity. We evaluate on five datasets with diverse sources, including image captions (SNLI), transcribed speech, popular fiction, and government reports (MNLI), Wikipedia articles (QNLI), science exams (SciTail) or news articles (RTE).

Table 2 details various results on the different NLI tasks for our model and previous state-of-the-art approaches. Our method significantly outperforms the baselines on four of the five datasets, achieving absolute improvements of upto 1.5% on MNLI, 5% on SciTail, 5.8% on QNLI and 0.6% on SNLI over the previous best results. This demonstrates our model’s ability to better reason over multiple sentences, and handle aspects of linguistic ambiguity. On RTE, one of the smaller datasets we evaluate on (2490 examples), we achieve an accuracy of 56%, which is below the 61.7% reported by a multi-task biLSTM model. Given the strong performance of our approach on larger NLI datasets, it is likely our model will benefit from multi-task training as well but we have not explored this currently.

²<https://ftfy.readthedocs.io/en/latest/>

³<https://spacy.io/>

表1：我们实验中使用的不同任务和数据集列表。

Task	Datasets
Natural language inference	SNLI [5], MultiNLI [66], Question NLI [64], RTE [4], SciTail [25]
Question Answering	RACE [30], Story Cloze [40]
Sentence similarity	MSR Paraphrase Corpus [14], Quora Question Pairs [9], STS Benchmark [6]
Classification	Stanford Sentiment Treebank-2 [54], CoLA [65]

但它在句子级别被打乱——破坏了长距离结构。我们的语言模型在该语料库上实现了极低的词元级别困惑度，仅为18.4。

模型规格 我们的模型主要遵循原始Transformer工作[62]。我们训练了一个12层的仅解码器Transformer，带有掩码自注意力头（768维状态和12个注意力头）。对于逐位置前馈网络，我们使用了3072维内部状态。我们采用了Adam优化方案[27]，最大学习率为2.5e-4。学习率在前2000次更新中从零线性增加，并通过余弦调度衰减至0。我们在小批量数据上训练了100个周期，每个小批量包含64个随机采样、连续的512个标记序列。由于模型中广泛使用了层归一化[2]，简单的权重初始化 $N(0, 0.02)$ 就足够了。我们使用了包含40,000次合并的字节对编码（BPE）词汇表[53]，以及残差、嵌入和注意力丢弃，丢弃率为0.1以进行正则化。我们还采用了[37]中提出的改进版L2正则化，对所有非偏置或增益权重应用 $w = 0.01$ 。对于激活函数，我们使用了高斯误差线性单元（GELU）[18]。我们使用了学习得到的位置嵌入，而不是原始工作中提出的正弦版本。我们使用*ftfy*库²来清理BooksCorpus中的原始文本，标准化一些标点和空格，并使用*spaCy*分词器³。

微调细节 除非另有说明，我们复用无监督预训练的超参数设置。我们在分类器中添加了丢弃率为0.1的dropout。对于大多数任务，我们使用6.25e-5的学习率和32的批次大小。我们的模型微调速度很快，3个训练周期对大多数情况已经足够。我们采用带预热的学习率线性衰减策略，预热阶段占训练总步数的0.2%。 λ 设置为0.5。

4.2 监督微调

我们在多种监督任务上进行了实验，包括自然语言推理、问答、语义相似性和文本分类。其中一些任务作为最近发布的GLUE多任务基准测试[64]的一部分提供，我们对此进行了利用。图1概述了所有任务和数据集。

自然语言推理 自然语言推理(NLI) 任务，也称为文本蕴含识别，涉及阅读一对句子并判断它们之间的关系属于蕴含、矛盾或中立中的一种。尽管最近引起了广泛关注[58, 35, 44]，但由于存在词汇蕴含、共指以及词汇和句法歧义等多种现象，该任务仍然具有挑战性。我们在五个来源多样的数据集上进行了评估，包括图像字幕（SNLI）、转录语音、流行小说和政府报告（MNLI）、维基百科文章（QNLI）、科学考试（SciTail）或新闻文章（RTE）。

表2详细列出了我们的模型与先前最先进方法在不同NLI任务上的各项结果。我们的方法在五个数据集集中的四个上显著超越基线，相较于先前最佳结果，在MNLI上实现了最高1.5%的绝对提升，在SciTail上提升5%，在QNLI上提升5.8%，在SNLI上提升0.6%。这证明了我们的模型能更好地进行多语句推理，并处理语言歧义问题。在RTE（我们评估的较小数据集之一，含2490个样本）上，我们取得了56%的准确率，低于多任务biLSTM模型报告的61.7%。鉴于我们的方法在大型NLI数据集上的优异表现，我们的模型很可能也能从多任务训练中受益，但目前尚未对此展开探索。

²<https://ftfy.readthedocs.io/en/latest/>

³<https://spacy.io/>

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	<u>77.6</u>	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
Finetuned Transformer LM (ours)	86.5	62.9	57.4	59.0

Question answering and commonsense reasoning Another task that requires aspects of single and multi-sentence reasoning is question answering. We use the recently released RACE dataset [30], consisting of English passages with associated questions from middle and high school exams. This corpus has been shown to contain more reasoning type questions that other datasets like CNN [19] or SQuAD [47], providing the perfect evaluation for our model which is trained to handle long-range contexts. In addition, we evaluate on the Story Cloze Test [40], which involves selecting the correct ending to multi-sentence stories from two options. On these tasks, our model again outperforms the previous best results by significant margins - up to 8.9% on Story Cloze, and 5.7% overall on RACE. This demonstrates the ability of our model to handle long-range contexts effectively.

Semantic Similarity Semantic similarity (or paraphrase detection) tasks involve predicting whether two sentences are semantically equivalent or not. The challenges lie in recognizing rephrasing of concepts, understanding negation, and handling syntactic ambiguity. We use three datasets for this task – the Microsoft Paraphrase corpus (MRPC) [14] (collected from news sources), the Quora Question Pairs (QQP) dataset [9], and the Semantic Textual Similarity benchmark (STS-B) [6]. We obtain state-of-the-art results on two of the three semantic similarity tasks (Table 4) with a 1 point absolute gain on STS-B. The performance delta on QQP is significant, with a 4.2% absolute improvement over Single-task BiLSTM + ELMo + Attn.

Classification Finally, we also evaluate on two different text classification tasks. The Corpus of Linguistic Acceptability (CoLA) [65] contains expert judgements on whether a sentence is grammatical or not, and tests the innate linguistic bias of trained models. The Stanford Sentiment Treebank (SST-2) [54], on the other hand, is a standard binary classification task. Our model obtains an score of 45.4 on CoLA, which is an especially big jump over the previous best result of 35.0, showcasing the innate linguistic bias learned by our model. The model also achieves 91.3% accuracy on SST-2, which is competitive with the state-of-the-art results. We also achieve an overall score of 72.8 on the GLUE benchmark, which is significantly better than the previous best of 68.9.

表2：自然语言推理任务的实验结果，将我们的模型与当前最先进方法进行比较。5x表示5个模型的集成。所有数据集均使用准确率作为评估指标。

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

表3：问答与常识推理结果，将我们的模型与当前最先进方法进行比较。9x表示9个模型的集成。

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	<u>77.6</u>	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
Finetuned Transformer LM (ours)	86.5	62.9	57.4	59.0

问答与常识推理 另一项需要单句和多句推理能力的任务是问答。我们使用最近发布的RACE数据集[30]，该数据集包含来自中学和高中考试的英文文章及相关问题。该语料库已被证明比其他数据集（如CNN[19]或SQuAD[47]）包含更多推理类问题，为我们经过训练以处理长距离上下文的模型提供了完美的评估。此外，我们在Story Cloze测试[40]上进行了评估，该测试涉及从两个选项中选择多句故事的正确答案。在这些任务中，我们的模型再次以显著优势超越了先前的最佳结果——在Story Cloze上提升了8.9%，在RACE上整体提升了5.7%。这证明了我们的模型有效处理长距离上下文的能力。

语义相似性 语义相似性（或复述检测）任务涉及预测两个句子在语义上是否等价。挑战在于识别概念的重述、理解否定以及处理句法歧义。我们为此任务使用了三个数据集——微软复述语料库（MRPC）[14]（从新闻来源收集）、Quora问题对（QQP）数据集[9]以及语义文本相似性基准（STS-B）[6]。我们在三个语义相似性任务中的两个上取得了最先进的结果（表4），其中STS-B的绝对增益为1分。QQP上的性能差异显著，相比单任务BiLSTM + ELMo + Attn，绝对提升了4.2%。

分类 最后，我们还在两个不同的文本分类任务上进行了评估。语言可接受性语料库CoLA [65]包含了专家对句子是否合乎语法的判断，用于测试训练模型的内在语言偏置。而斯坦福情感树库（SST-2）[54]则是一个标准的二分类任务。我们的模型在CoLA上获得了45.4的分数，相比之前的最佳结果35.0实现了显著提升，展示了模型学习到的内在语言偏置。该模型在SST-2上也达到了91.3%的准确率，与当前最先进的结果具有竞争力。此外，我们在GLUE基准测试中获得了72.8的总分，显著优于之前68.9的最佳成绩。

Table 4: Semantic similarity and classification results, comparing our model with current state-of-the-art methods. All task evaluations in this table were done using the GLUE benchmark. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

Method	Classification		Semantic Similarity			GLUE
	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	
Sparse byte mLSTM [16]	-	93.2	-	-	-	-
TF-KLD [23]	-	-	86.0	-	-	-
ECNU (mixed ensemble) [60]	-	-	-	81.0	-	-
Single-task BiLSTM + ELMo + Attn [64]	35.0	90.2	80.2	55.5	66.1	64.8
Multi-task BiLSTM + ELMo + Attn [64]	18.9	91.6	83.5	72.8	63.3	68.9
Finetuned Transformer LM (ours)	45.4	91.3	82.3	82.0	70.3	72.8

Overall, our approach achieves new state-of-the-art results in 9 out of the 12 datasets we evaluate on, outperforming ensembles in many cases. Our results also indicate that our approach works well across datasets of different sizes, from smaller datasets such as STS-B ($\approx 5.7k$ training examples) – to the largest one – SNLI ($\approx 550k$ training examples).

5 Analysis

Impact of number of layers transferred We observed the impact of transferring a variable number of layers from unsupervised pre-training to the supervised target task. Figure 2(left) illustrates the performance of our approach on MultiNLI and RACE as a function of the number of layers transferred. We observe the standard result that transferring embeddings improves performance and that each transformer layer provides further benefits up to 9% for full transfer on MultiNLI. This indicates that each layer in the pre-trained model contains useful functionality for solving target tasks.

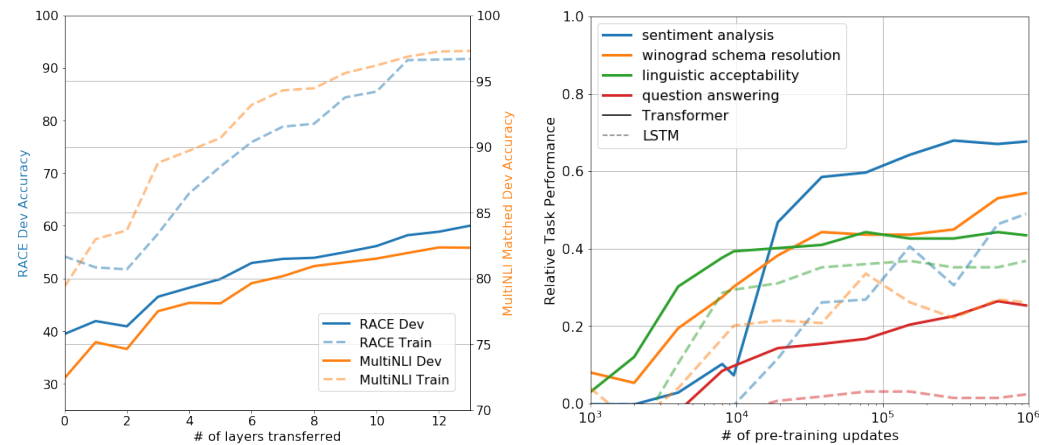


Figure 2: (left) Effect of transferring increasing number of layers from the pre-trained language model on RACE and MultiNLI. (right) Plot showing the evolution of zero-shot performance on different tasks as a function of LM pre-training updates. Performance per task is normalized between a random guess baseline and the current state-of-the-art with a single model.

Zero-shot Behaviors We’d like to better understand why language model pre-training of transformers is effective. A hypothesis is that the underlying generative model learns to perform many of the tasks we evaluate on in order to improve its language modeling capability and that the more structured

表4：语义相似性与分类结果，将我们的模型与当前最先进方法进行比较。本表中所有任务评估均使用GLUE基准进行。（*mc*=马修斯相关系数，*acc*=准确率，*pc*=皮尔逊相关系数）

Method	Classification		Semantic Similarity			GLUE
	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	
Sparse byte mLSTM [16]	-	93.2	-	-	-	-
TF-KLD [23]	-	-	86.0	-	-	-
ECNU (mixed ensemble) [60]	-	-	-	81.0	-	-
Single-task BiLSTM + ELMo + Attn [64]	35.0	90.2	80.2	55.5	66.1	64.8
Multi-task BiLSTM + ELMo + Attn [64]	18.9	91.6	83.5	72.8	63.3	68.9
Finetuned Transformer LM (ours)	45.4	91.3	82.3	82.0	70.3	72.8

总体而言，我们的方法在评估的12个数据集中有9个取得了新的最先进成果，在许多情况下超越了集成方法。我们的结果还表明，该方法在不同规模的数据集上均表现良好，从较小的数据集如STS-B ($\approx 5.7k$ 训练样本) 到最大的数据集SNLI ($\approx 550k$ 训练样本) 皆是如此。

5 分析

迁移层数的影响 我们观察了将不同层数从无监督预训练迁移到有监督目标任务的影响。图（左）展示了我们的方法在MultiNLI和RACE任务上性能随迁移层数变化的曲线。我们观察到标准结果：迁移词嵌入能提升性能，而每个Transformer层的迁移会带来额外增益——在MultiNLI任务上完全迁移所有层可获得高达9%的性能提升。这表明预训练模型的每一层都包含解决目标任务所需的有效功能。

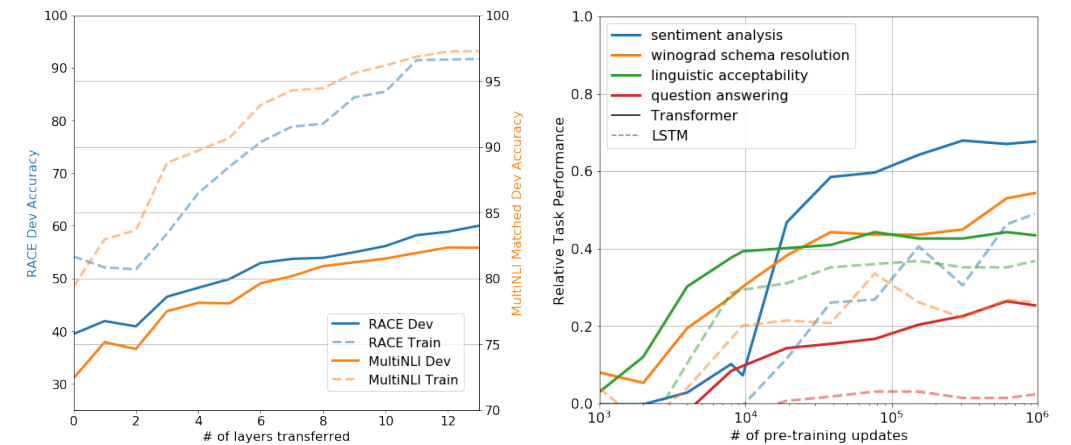


图2：（左）从预训练语言模型中迁移递增层数对RACE和MultiNLI的影响。（右）展示不同任务上零样本性能随语言模型预训练更新次数变化的曲线图。每个任务的性能在随机猜测基线当前单模型最优性能之间进行了归一化处理。

零样本行为 我们想更好地理解为什么transformer的语言模型预训练是有效的。一个假设是，底层生成模型通过学习执行我们评估的许多任务来提高其语言建模能力，并且结构越

Table 5: Analysis of various model ablations on different tasks. Avg. score is a unweighted average of all the results. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

Method	Avg. Score	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	MNLI (acc)	QNLI (acc)	RTE (acc)
Transformer w/ aux LM (full)	74.7	45.4	91.3	82.3	82.0	70.3	81.8	88.1	56.0
Transformer w/o pre-training	59.9	18.9	84.0	79.4	30.9	65.5	75.7	71.2	53.8
Transformer w/o aux LM	75.0	47.9	92.0	84.9	83.2	69.8	81.1	86.9	54.4
LSTM w/ aux LM	69.1	30.3	90.5	83.2	71.8	68.1	73.7	81.1	54.6

attentional memory of the transformer assists in transfer compared to LSTMs. We designed a series of heuristic solutions that use the underlying generative model to perform tasks without supervised finetuning. We visualize the effectiveness of these heuristic solutions over the course of generative pre-training in Fig 2(right). We observe the performance of these heuristics is stable and steadily increases over training suggesting that generative pretraining supports the learning of a wide variety of task relevant functionality. We also observe the LSTM exhibits higher variance in its zero-shot performance suggesting that the inductive bias of the Transformer architecture assists in transfer.

For CoLA (linguistic acceptability), examples are scored as the average token log-probability the generative model assigns and predictions are made by thresholding. For SST-2 (sentiment analysis), we append the token *very* to each example and restrict the language model’s output distribution to only the words *positive* and *negative* and guess the token it assigns higher probability to as the prediction. For RACE (question answering), we pick the answer the generative model assigns the highest average token log-probability when conditioned on the document and question. For DPRD [46] (winograd schemas), we replace the definite pronoun with the two possible referrents and predict the resolution that the generative model assigns higher average token log-probability to the rest of the sequence after the substitution.

Ablation studies We perform three different ablation studies (Table 5). First, we examine the performance of our method without the auxiliary LM objective during fine-tuning. We observe that the auxiliary objective helps on the NLI tasks and QQP. Overall, the trend suggests that larger datasets benefit from the auxiliary objective but smaller datasets do not. Second, we analyze the effect of the Transformer by comparing it with a single layer 2048 unit LSTM using the same framework. We observe a 5.6 average score drop when using the LSTM instead of the Transformer. The LSTM only outperforms the Transformer on one dataset – MRPC. Finally, we also compare with our transformer architecture directly trained on supervised target tasks, without pre-training. We observe that the lack of pre-training hurts performance across all the tasks, resulting in a 14.8% decrease compared to our full model.

6 Conclusion

We introduced a framework for achieving strong natural language understanding with a single task-agnostic model through generative pre-training and discriminative fine-tuning. By pre-training on a diverse corpus with long stretches of contiguous text our model acquires significant world knowledge and ability to process long-range dependencies which are then successfully transferred to solving discriminative tasks such as question answering, semantic similarity assessment, entailment determination, and text classification, improving the state of the art on 9 of the 12 datasets we study. Using unsupervised (pre-)training to boost performance on discriminative tasks has long been an important goal of Machine Learning research. Our work suggests that achieving significant performance gains is indeed possible, and offers hints as to what models (Transformers) and data sets (text with long range dependencies) work best with this approach. We hope that this will help enable new research into unsupervised learning, for both natural language understanding and other domains, further improving our understanding of how and when unsupervised learning works.

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表5：不同任务上各种模型消融分析。平均分是所有结果的未加权平均值。（*mc*=马修斯相关系数，*acc*=准确率，*pc*=皮尔逊相关系数）

Method	Avg. Score	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	MNLI (acc)	QNLI (acc)	RTE (acc)
Transformer w/ aux LM (full)	74.7	45.4	91.3	82.3	82.0	70.3	81.8	88.1	56.0
Transformer w/o pre-training	59.9	18.9	84.0	79.4	30.9	65.5	75.7	71.2	53.8
Transformer w/o aux LM	75.0	47.9	92.0	84.9	83.2	69.8	81.1	86.9	54.4
LSTM w/ aux LM	69.1	30.3	90.5	83.2	71.8	68.1	73.7	81.1	54.6

与LSTM相比，Transformer的注意力记忆机制有助于迁移学习。我们设计了一系列启发式解决方案，这些方案利用底层生成模型来执行任务，而无需监督微调。在图2（右）中，我们可视化了这些启发式解决方案在生成式预训练过程中的有效性。我们观察到这些启发式的性能稳定，并随着训练稳步提升，这表明生成式预训练支持学习多种任务相关功能。我们还观察到LSTM在零样本性能上表现出更高的方差，这表明Transformer架构的归纳偏置有助于迁移。

对于CoLA（语言可接受性），示例的评分是生成模型分配的平均标记对数概率，并通过阈值处理进行预测。对于SST-2（情感分析），我们在每个示例后附加标记*very*，并将语言模型的输出分布限制在仅包含词语*positive*和*negative*的范围内，然后猜测模型赋予更高概率的标记作为预测。对于RACE（问答），我们选择生成模型在给定文档和问题条件下分配最高平均标记对数概率的答案。对于DPRD [46]（威诺格拉德模式），我们用两个可能的指代替换定代词，并预测生成模型在替换后对序列其余部分赋予更高平均标记对数概率的解析结果。

消融研究我们进行了三项不同的消融研究（表5）。首先，我们检验了在微调过程中不使用辅助语言模型目标时我们方法的性能。我们观察到辅助目标在NLI任务和QQP上有所帮助。总体趋势表明，较大的数据集从辅助目标中受益，而较小的数据集则不然。其次，我们通过将Transformer与使用相同框架的单层2048单元LSTM进行比较，分析了Transformer的效果。当使用LSTM代替Transformer时，我们观察到平均分数下降了5.6分。LSTM仅在一个数据集——MRPC上表现优于Transformer。最后，我们还与直接在监督目标任务上训练（未经预训练）的Transformer架构进行了比较。我们观察到，缺乏预训练会损害所有任务的性能，导致与我们的完整模型相比性能下降了14.8%。

6 结论

我们引入了一个框架，通过生成式预训练和判别式微调，实现了一个单一任务无关模型的强大自然语言理解能力。通过在包含长连续文本的多样化语料库上进行预训练，我们的模型获得了重要的世界知识和处理长距离依赖的能力，这些能力随后被成功迁移到解决判别式任务上，如问答、语义相似性评估、蕴含判定和文本分类，从而在我们研究的12个数据集集中的9个上提升了现有技术水平。利用无监督（预）训练来提升判别式任务的性能，长期以来一直是机器学习研究的重要目标。我们的工作表明，实现显著的性能提升确实是可能的，并提供了关于哪些模型（Transformer）和数据集（具有长距离依赖的文本）在这种方法中效果最佳的线索。我们希望这将有助于推动无监督学习的新研究，无论是在自然语言理解还是其他领域，进一步增进我们对无监督学习如何及何时有效的理解。

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